

Anton Ernstsson

Junior Data Scientist / Applied AI Developer

Python | Machine Learning | NLP | Data Engineering

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Swedish (native) | English (fluent)

PROFILE

Junior AI developer with a background in data engineering and applied machine learning. My experience includes analysing real-world support data to uncover recurring patterns and potential problems. I am particularly interested in unsupervised learning and enjoy exploring how data quality and context shape the insights we can draw.

TECHNICAL SKILLS

Languages and data: Python, SQL, pandas, NumPy, PySpark, Azure Databricks

Machine learning: scikit-learn, XGBoost, Random Forest, regression, classification, clustering, feature engineering

NLP and generative AI: sentence-transformers, UMAP, HDBSCAN, RAG, embeddings, LLM APIs

Deep learning: TensorFlow, Keras, CNN, MLP, image and audio classification

Evaluation: Cross-validation, ROC-AUC, F1, confusion matrix, hypothesis testing, statistical analysis

MLOps and engineering: MLflow, Git, pytest, CI/CD, REST APIs, data validation

PROFESSIONAL EXPERIENCE

Analytics Engineer | Avtalat

Jan 2026 - Jun 2026

LIA internship

- Developed a GDPR-compliant NLP solution that clustered 21,000+ production incidents and identified 72 potential unregistered problem records.
- Compared clustering approaches and evaluated how ISO/IEC 25012 data quality and process maturity affected downstream model performance.
- Extended the underlying Databricks ETL pipeline and built Power BI semantic models for business users.

Data Engineer | Fora AB

Nov 2025 - Jan 2026

LIA internship

- Built Python and PySpark pipelines ingesting 80,000+ records into a Bronze, Silver and Gold lakehouse architecture.
- Implemented validation and data modelling that created a reliable analytical foundation for reporting and AI use cases.
- Worked with sensitive operational data in a regulated pension and insurance environment.

EDUCATION

AI Developer | JENSEN Yrkeshögskola

2024 - 2026

Higher Vocational Education (YH), 400 credits, two years full-time. Graduated June 2026.

ADDITIONAL EXPERIENCE

Machine Operator | Delicato AB

2022 - 2025

Production, quality control and technical troubleshooting. Vice-chair of the local union club.

Selected projects

Machine learning, NLP, deep learning and reproducible analysis

NLP Clustering of ITSM Data

Graduation thesis

- Built an unsupervised NLP pipeline on 21,000+ real production incidents, identifying 72 potential problem records and outperforming a K-means baseline on cluster quality.
- Combined quantitative model evaluation with an ISO/IEC 25012 data-quality assessment and qualitative stakeholder interviews.

Python | sentence-transformers | UMAP | HDBSCAN | K-means | MLflow | Databricks

Industrial Pump Anomaly Detection

Deep learning on the MIMII dataset

- Converted raw pump audio into Mel spectrograms and trained a CNN-based binary classifier for industrial anomaly detection.
- Evaluated performance with ROC-AUC, confusion matrix and classification reports while controlling overfitting with Dropout and L2 regularisation.

Python | TensorFlow | CNN | librosa | signal processing

Drug Synergy Prediction

Multimodal machine learning

- Combined drug fingerprints, RNA expression and cell-line metadata to predict synergy, with interaction-aware feature engineering and leakage prevention.
- Compared ensemble methods and deep learning with dimensionality reduction and structured model evaluation.

Python | scikit-learn | TensorFlow | Random Forest | Gradient Boosting | UMAP

RAG Learning Assistant

Applied generative AI

- Built a document-based application supporting embeddings, semantic search, grounded answers and quiz generation.

Python | Flask | Vertex AI | Google Cloud Storage | RAG